CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 51 (2015) 24–34

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

A thermodynamic description of the Al–B–Er system

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ARTICLE INFO

Article history:
Received 14 January 2015
Received in revised form
12 July 2015
Accepted 14 July 2015
Available online 24 July 2015

Keywords:
Al–B–Er system
CALPHAD technique
Phase equilibria
Thermodynamic properties

ABSTRACT

The Al–B–Er ternary system was optimized by means of the CALPHAD (CALculation of PHAse Diagram) technique. The solution phases, liquid and hcp, were described by the substitutional solution model. The compounds AlB_{12}, AlB_2, B_2Er, B_4Er, $B_{12}Er$, $B_{66}Er$, AlB_4Er and $AlB_{14}Er$ were treated as stoichiometric compounds. In the present study, in order to describe the structural relationship between the solution phase fcc(A1) (disordered part) and the phase $L1_2$ (*cP*4-$AuCu_3$-type, ordered part), the two phases were modeled as $(Al,B,Er)_1(Va)_1$ and $(Al,B,Er)_{0.75}(Al,B,Er)_{0.25}(Va)_1$, respectively. A self-consistent thermodynamic description of the Al–B–Er system was obtained.

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1. Introduction

Three types of ternary compounds, RE_2AlB_6, $REAlB_4$, and $REAlB_{14}$, have been found in RE–Al–B systems [1–6]. In particular, $REAlB_4$ compounds are of great interest because of their remarkable magnetic properties, which in many cases are of potential interest for applications towards thermoelectric and photodetector materials [7] and heavy fermion superconductors [8]. So a reliable thermodynamic description and related theoretical analysis of these systems is quite useful for the design of materials.

In order to assess the Al–B–Er ternary system, the interaction parameters of the liquid phase and the fcc(A1) phase in the Al–B system were taken from the study of Mirković et al. [9] and the interaction parameters of the hcp phase in the B–Er system were taken from the study of Wang et al. [10], while the other thermodynamic parameters in the Al–B and the B–Er systems were re-optimized. The thermodynamic description of the Al–Er system, which was adopted in this study, was assessed by Cacciamani [11]. Fig. 1 represents the calculated Al–Er phase diagram from Cacciamani's study [11], and Table 1 lists the experimental and calculated results for the Al–Er phase, using experimental data taken from the literature [11,22,24].

Here, the thermodynamic description of the Al–B–Er system was done using the CALPHAD technique. A set of self-consistent parameters for the calculation of the phase equilibria and the thermochemical properties of the system are provided.

2. Literature information

2.1. The binary system

Serebryanskii et al. [12] investigated the liquidus curve and considered the phases β-AlB_{12} and AlB_{10} as stable. However, the phase diagram in Serebryanskii et al. [12] contradicts the phase rule in the boron-rich region because there is no two-phase region between AlB_{12} and B(rhom). Later, Samsonov et al. [13] and Duschanek and Rogl [14] concluded that the compounds β-AlB_{12} and AlB_{10} were stabilized by impurities. Duschanek and Rogl [14] considered the phase γ-AlB_{12}, which was found in the studies of Kohn et al. [15] and Higashi and Takahashi [16], to also be stable. Mirković et al. [9] considered the compound Al_2B_3, which was found in the study of Vardiman [17], as unstable. Mirković et al. [9] and Grytsiv and Rogl [18] confirmed two stable compounds, AlB_2 and AlB_{12}, and modified the phase equilibria accordingly. Several other thermodynamic assessments of the Al–B system can be found in different studies (e.g. [9,14,19–21]).

Buschow and Vucht [22] determined the phase equilibria of the Al–Er system, Gschneidner and Calderwood [23] suggested that the solubility of Er in Al is negligible, and Saccone et al. [24] investigated the liquidus curve in the composition range of 50–80 at% Er. Many researchers [11, 23, 25–27] have assessed the Al–Er system. Cacciamani et al. [11] modeled the phase $L1_2$ (*cP*4-AuCu3-type) as the ordered phase of fcc(A1). Jin et al. [27] optimized the liquid phase using the modified quasichemical model; hence, their assessment was not used in this study.

The phase diagram of the B–Er system has also been investigated by many researchers [28–31]. The standard enthalpy of formation of B_2Er was measured using direct high-temperature

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http://dx.doi.org/10.1016/j.calphad.2015.07.001
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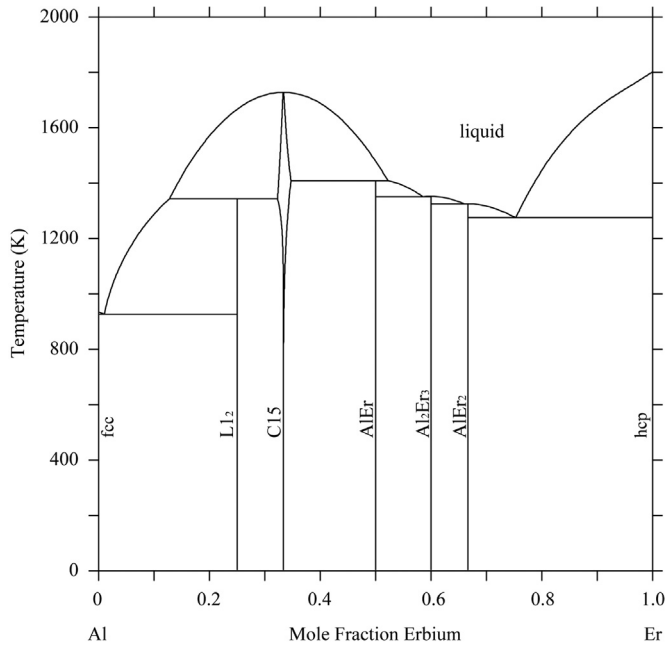

Fig. 1. Al–Er phase diagram, plotted using data from Cacciamani [11].

reaction calorimetry at 1473 K by Meschel and Kleppa [32], and was earlier predicted by the same authors [33]. Wang et al. [10] assessed the B–Er system; in their study, all the four intermetallic compounds, namely B_2Er, B_4Er, $B_{12}Er$, and $B_{66}Er$, were treated as stoichiometric compounds, with the temperature of the miscibility gap having a value of ~4148 K.

2.2. The Al–B–Er ternary system

In the Al–B–Er ternary system, two ternary compounds are known: AlB_4Er [34,35] ($YCrB_4$-type orthorhombic, denoted as τ_1 here) and $AlB_{14}Er$ [35] ($MgAlB_{14}$-type orthorhombic, denoted as τ_2 here).

Yu and Lundström [36] found that the ternary compound τ_2, a metal-deficient compound, was represented accurately by the formula $Al_{1-y}B_{14}Er_{1-x}$ within a narrow homogeneity range around $Al_{0.62}B_{14}Er_{0.57}$. They investigated the B-rich region (in the composition range of ~66–100 at% B) of the isothermal section at 1873 K by X-ray powder diffraction and electron probe microanalysis and found that the composition range of τ_2 was $Al_{0.64}B_{14}Er_{0.55}$–$Al_{0.58}B_{14}Er_{0.58}$.

The partial isothermal section at 1073 K was determined using X-ray powder diffraction by Chaban et al. [35]. They also investigated the ternary compound τ_2, and found that it was slightly more B-rich than is indicated by the nominal formula $AlB_{14}Er$. However, they did not draw the phase equilibria in most of the composition region $Al_xB_{1-x}Er_0$– $Al_xB_{0.95-x}Er_{0.05}$.

Petsow and Effenberg [37] evaluated the isothermal section at 873 K in the composition range of 0–75 at% B in their 1990 study. They did not find τ_1, τ_2 or other ternary compounds.

3. Thermodynamic models

3.1. Unary phases

The Gibbs energy functions of the unary phases of elements Al, B and Er were taken from the SGTE (Scientific Group Thermodata Europe) pure elements database compiled by Dinsdale [38]. The phase rhom (βB) was described by the $(B)_{0.8857}(B)_{0.1143}$ model, which is the same as the model used by Mirković et al. [9].

3.2. Solution phases

In the Al–B–Er system, there are three solution phases: liquid, fcc(A1), and hcp(A3), which were modeled by the substitutional solution model. Their Gibbs energies are given by

$$G_m^\varphi = x_{Al}G_{Al}^\varphi(T) + x_B G_B^\varphi(T) + x_{Er}G_{Er}^\varphi(T) + RT(x_{Al}\ln x_{Al} + x_B \ln x_B + x_{Er}\ln x_{Er}) + {}^E G_m^\varphi \quad (1)$$

where R is the ideal gas constant, x_i (i=Al, B, Er) is the mole fraction of the pure element i, and ${}^E G_m^\varphi$ is the excess Gibbs energy expressed by the Redlich–Kister–Muggianu formalism [39,40]:

$$\begin{aligned} {}^E G_m^\varphi = & x_{Al}x_B \sum_n {}^n L_{Al,B}^\varphi (x_{Al} - x_B)^n + x_{Al}x_{Er} \sum_n {}^n L_{Al,Er}^\varphi (x_{Al} - x_{Er})^n \\ & + x_B x_{Er} \sum_n {}^n L_{B,Er}^\varphi (x_B - x_{Er})^n \\ & + x_{Al}x_B x_{Er} L_{Al,B,Er}^\varphi \end{aligned} \quad (2)$$

where ${}^n L_{i,j}^\varphi$ (i, j=Al, B, Er, and i≠j) are the nth binary interaction parameters between elements i and j, and their general form is

$$L^\varphi = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} \quad (3)$$

In most cases, only the first one or two terms on the right-hand side in Eq. (3) (i.e. "a" or "$a+bT$" in) are used, according to the temperature dependence of the experimental data. $L_{Al,B,Er}^\varphi$ is the ternary interaction parameter and is expressed as

$$L_{Al,B,Er}^\varphi = x_{Al}{}^0 L_{Al,B,Er}^\varphi + x_B{}^1 L_{Al,B,Er}^\varphi + x_{Er}{}^2 L_{Al,B,Er}^\varphi \quad (4)$$

where ${}^n L_{Al,B,Er}^\varphi = a_n^\varphi + b_n^\varphi T$ (n=0, 1, 2), and a_n^φ and b_n^φ are the parameters to be optimized in this study.

3.3. Stoichiometric intermetallic compounds

The compounds AlB_{12}, AlB_2, B_2Er, B_4Er, $B_{12}Er$ and $B_{66}Er$ were

Table 1
Invariant reactions of the Al–Er system.

Reaction	Type	Experimental data [22,24]				Calculated results [11]			
		T(K)		x(Er)		T(K)		x(Er)	
liq.→$L1_2$+fcc	Eutectic	928[22]	0.016	0.250	0.001	921	0.0170	0.250	0.001
liq.+C15→$L1_2$	Peritectic	1343[22]	0.160	0.333	0.250	1235	0.0764	0.312	0.250
liq.→C15	Congruent	1728[22]	0.333	0.333	–	1722	0.333	0.333	–
liq.+C15→AlEr	Peritectic	1413[24]	0.524	0.333	–	1404	0.525	0.350	–
liq.→AlEr+Al_2Er_3	Eutectic	1343[24]	0.575	–	–	1353	0.578	–	–
liq.→Al_2Er_3	Congruent	1358[24]	0.600	–	–	1357	0.600	–	–
liq.→Al_2Er_3+$AlEr_2$	Eutectic	1313[24]	0.650	–	–	1329	0.657	–	–
liq.→$AlEr_2$	Congruent	1330[24]	0.667	–	–	1330	0.667	–	–
liq.→hcp+$AlEr_2$	Eutectic	1278[24]	0.750	0.999	–	1273	0.753	0.999	–

Table 2
Thermodynamic parameters of the Al–B–Er system[a].

Phase	Thermodynamic parameters		Reference
Liquid	Model $(Al,B,Er)_1$		
	${}^0L^{liq.}_{Al,B} = -927.5300 - 5.1131 \cdot T$		[9]
	${}^1L^{liq.}_{Al,B} = +31961.6$		[9]
	${}^2L^{liq.}_{Al,B} = -12992.0$		[9]
	${}^0L^{liq.}_{Al,Er} = -176486.0 + 55.6852 \cdot T$		[11]
	${}^1L^{liq.}_{Al,Er} = -36685.5 + 23.4492 \cdot T$		[11]
	${}^2L^{liq.}_{Al,Er} = +34349.1 - 8.2352 \cdot T$		[11]
	${}^0L^{liq.}_{B,Er} = -91952.5 + 9.6936 \cdot T$		This work
	${}^1L^{liq.}_{B,Er} = -72534.4 + 6.6000 \cdot T$		This work
	${}^2L^{liq.}_{B,Er} = -27945.3 - 0.4828 \cdot T$		This work
	${}^3L^{liq.}_{B,Er} = -9.9973 \cdot T$		This work
fcc(A1)	Model $(Al,B,Er)_1(Va)_1$		
	$G(fcc(A1), Al{:}Va) = GHSER_{Al}$	$(298\ K < T < 6000\ K)$	[9]
	$G(fcc(A1), B{:}Va) = GHSER_B + 50208.0 - 13.4780 \cdot T$	$(298\ K < T < 6000\ K)$	[9]
	$G(fcc(A1), Er{:}Va) = GHSER_{Er} + 1950.0 - 0.2924 \cdot T$	$(298\ K < T < 6000\ K)$	[11]
	${}^0L^{fcc}_{Al,B:Va} = +21418.6 - 1.0559T$		This work
hcp	Model $(Al,B,Er)_1(Va)_1$		
	$G(hcp, Al{:}Va) = GHSER_{Al} + 5481.0 - 1.8000 \cdot T$	$(298\ K < T < 6000\ K)$	[11]
	$G(hcp, B{:}Va) = 0$		This work
	$G(hcp, Er{:}Va) = GHSER_{Er}$	$(298\ K < T < 3200\ K)$	[11]
	${}^0G^{hcp}_{B,Er:Va} = 50000.0$		[10]
AlB_2	Model $(Al)_{0.3333}(B)_{0.6667}$		
	${}^0G^{AlB2}_{Al:B} = 0.3333 \times GHSER_{Al} + 0.6667 \times GHSER_B - 6588.4 - 1.4732 \cdot T$		This work
AlB_{12}	Model $(Al)_{0.0769}(B)_{0.9231}$		
	${}^0G^{AlB2}_{Al:B} = 0.0769 \times GHSER_{Al} + 0.9231 \times GHSER_B - 10025.8 - 0.1843 \cdot T$		This work
AlEr	Model $(Al)_{0.5}(Er)_{0.5}$		
	${}^0G^{AlB2}_{Al:B} = 0.5 \times GHSER_{Al} + 0.5 \times GHSER_B - 50000.0 + 11.9922 \cdot T$		This work
Al_2Er_3	Model $(Al)_{0.4}(Er)_{0.6}$		
	${}^0G^{AlB2}_{Al:B} = 0.4 \times GHSER_{Al} + 0.6 \times GHSER_B - 45000.0 + 10.8726 \cdot T$		This work
$AlEr_2$	Model $(Al)_{0.3333}(Er)_{0.6667}$		
	${}^0G^{AlB2}_{Al:B} = 0.3333 \times GHSER_{Al} + 0.6667 \times GHSER_B - 39000.0 + 9.2228 \cdot T$		This work
B_2Er	Model $(B)_{0.6667}(Er)_{0.3333}$		
	${}^0G^{B2Er}_{B:Er} = 0.6667 \times GHSER_B + 0.3333 \times GHSER_{Er} - 74100.0 + 14.9987 \cdot T$		This work
B_4Er	Model $(B)_{0.8}(Er)_{0.2}$		
	${}^0G^{B4Er}_{B:Er} = 0.8 \times GHSER_B + 0.2 \times GHSER_{Er} - 54025.0 + 6.8497 \cdot T$		This work
$B_{12}Er$	Model $(B)_{0.9231}(Er)_{0.0769}$		
	${}^0G^{B12Er}_{B:Er} = 0.9231 \times GHSER_B + 0.0769 \times GHSER_{Er} - 32138.3 + 6.5201 \cdot T$		This work
$B_{66}Er$	Model $(B)_{0.9848}(Er)_{0.0152}$		
	${}^0G^{B12Er}_{B:Er} = 0.9851 \times GHSER_B + 0.0149 \times GHSER_{Er} - 12422.1 + 3.5522 \cdot T$		This work
$L1_2$	Model $(Al,B,Er)_{0.75}(Al,B,Er)_{0.25}(Va)_1$		
	${}^0G^{L12}_{Al:Al:Va} = 0$		[11]
	${}^0G^{L12}_{Al:B:Va} = 0$		[11]

Table 2 (*continued*)

Phase	Thermodynamic parameters	Reference
	${}^{0}G^{L12}_{Al:Er:Va} = -88442.2 + 23.3163 \cdot T$	[11]
	${}^{0}G^{L12}_{B:Al:Va} = 0$	[11]
	${}^{0}G^{L12}_{B:B:Va} = 0$	[11]
	${}^{0}G^{L12}_{B:Er:Va} = 0$	[11]
	${}^{0}G^{L12}_{Er:Al:Va} = 0$	[11]
	${}^{0}G^{L12}_{Er:B:Va} = 0$	[11]
	${}^{0}G^{L12}_{Er:Er:Va} = 0$	[11]
	${}^{0}L^{L12}_{Al,Er:\ Al:Va} = -112887 + 27.7412 \cdot T$	[11]
	${}^{1}L^{L12}_{Al,Er:\ Al:Va} = -112887 + 27.7412 \cdot T$	[11]
	${}^{0}L^{L12}_{Al,Er:\ Er:Va} = +112887 - 27.7412 \cdot T$	[11]
	${}^{1}L^{L12}_{Al,Er:\ Er:Va} = +37629.0 - 9.2471 \cdot T$	[11]
C15	Model $(Al,Er)_2(Al,Er)_1$	
	${}^{0}G^{C15}_{Al:\ Al} = 3 \times GHSER_{Al} + 15000.0$	[11]
	${}^{0}G^{C15}_{Al:\ Er} = 2 \times GHSER_{Al} + GHSER_{Er} - 165000.0 + 40.3380 \cdot T$	[11]
	${}^{0}G^{C15}_{Er:\ Al} = GHSER_{Al} + 2 \times GHSER_{Er} + 165000.0 + 30000 - 40.3380 \cdot T$	[11]
	${}^{0}G^{C15}_{Er:\ Er} = 3 \times GHSER_{Er} + 15000.0$	[11]
τ_1 (AlB_4Er)	Model $(Al)_1(B)_4(Er)_1$	
	${}^{0}G^{\tau 1}_{Al:B:Er} = GHSER_{Al} + 4 \times GHSER_{B} + GHSER_{Er} - 270966.2 + 36.6675 \cdot T - 3.9776 \times 10^{-12} \cdot T^5$	This work
τ_2 ($AlB_{14}Er$)	model $(Al)_1(B)_{14}(Er)_1$	
	${}^{0}G^{\tau 2}_{Al:B:Er} = GHSER_{Al} + 14 \times GHSER_{B} + GHSER_{Er} - 455624.5 + 99.8871 \cdot T - 2.7806 \times 10^{-15} \cdot T^6$	This work

[a] In J/mol of the formula unit.

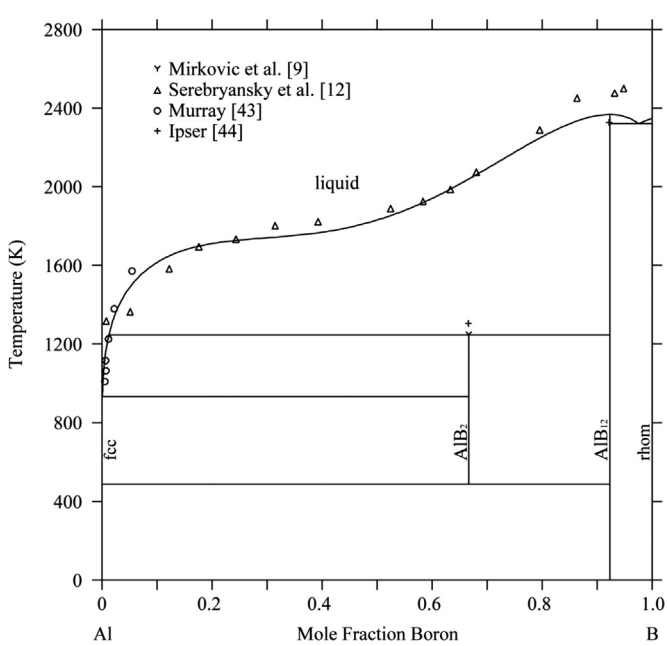

Fig. 2. Al–B phase diagram calculated using experimental data from the literature [9,12,43,44].

treated as stoichiometric compounds (denoted as M_xN_y) in the present study. The Gibbs energy per mole of formula unit M_xN_y is expressed as follows:

$$G_m^{M_xN_y} = xGHSER_M + yGHSER_N + \Delta G_f^{M_xN_y} \quad (5)$$

where $\Delta G_f^{M_xN_y}$ is the Gibbs energy of formation per mole of formula unit M_xN_y. The binary compounds AlB_{12}, AlB_2, AlEr, Al_2Er_3, $AlEr_2$, B_2Er, B_4Er, $B_{12}Er$ and $B_{66}Er$ in the ternary Al–B–Er system have no significant solubility of the third component at 1073 K [35] and 1873 K [36], so the thermodynamic models of the binary compounds in the ternary Al–B–Er system are consistent with the models in the Al–B, Al–Er and B–Er binary systems.

The Gibbs energies of the ternary compounds τ_1 and τ_2 per mole of formula unit are expressed as

$$G_m^{\tau 1} = GHSER_{Al} + 4GHSER_B + GHSER_{Er} + \Delta G_f^{\tau 1} \quad (6)$$

and

$$G_m^{\tau 2} = GHSER_{Al} + 14GHSER_B + GHSER_{Er} + \Delta G_f^{\tau 2} \quad (7)$$

where $\Delta G_f^{\tau 1}$ and $\Delta G_f^{\tau 2}$ are the Gibbs energies of formation per mole of formula unit τ_1 and τ_2, respectively.

3.4. Intermetallic compounds C15

The phase C15 was described by a two-sublattice model $(Al,Er)_2(Al,Er)_1$, with Al and Er in both the first and the second sublattices. The Gibbs energy per mole of formula unit $(Al,Er)_2(Al,Er)_1$ is described as

Table 3
Invariant reactions of the Al–B system.

Reaction	Type	Present work				Experimental data [9]			
		T(K)		x(B)		T(K)		x(B)	
liq.→AlB_{12}+rhom	Eutectic	2320	0.975	–	0.999	2320	0.975	–	0.999
liq.→AlB_{12}	Congruent	2367	0.923	–		2367	0.923	–	
liq.+AlB_{12}→AlB_2	Peritectic	1245	0.0125	–	–	1245	0.0120	–	–
liq.→AlB_2+fcc	Eutectic	933	8.65×10^{-4}	–	1.20×10^{-4}	933	7.60×10^{-4}	–	5.40×10^{-5}
AlB_2→fcc+AlB_2	Eutectic	487	–	5.40×10^{-6}	–	486	–	0.001	–

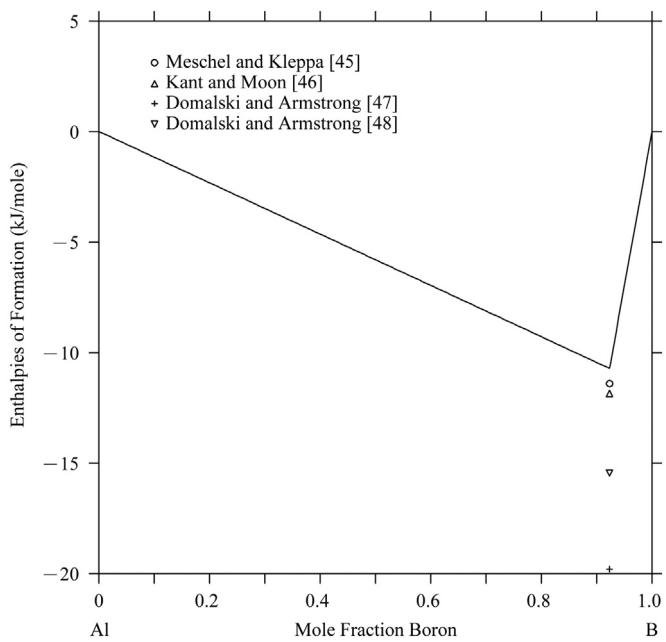

Fig. 3. Enthalpies of formation at 298 K in the Al–B system, calculated using experimental data from the literature [45–48].

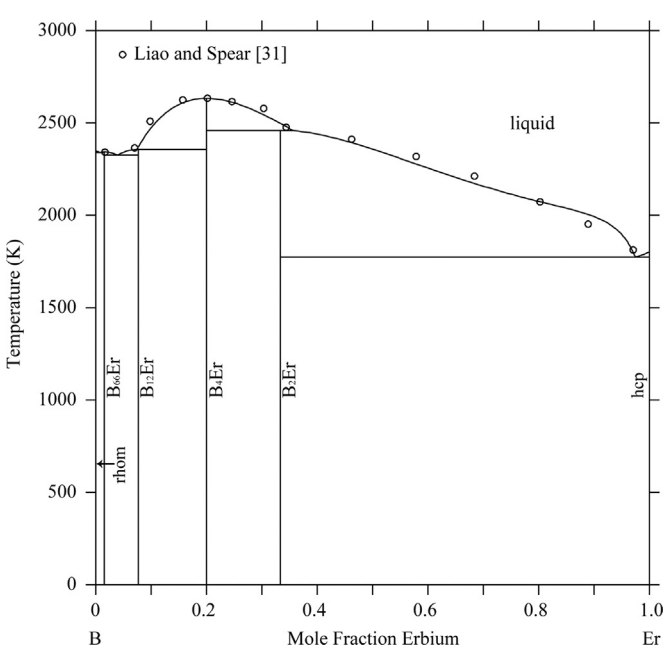

Fig. 4. B–Er phase diagram calculated using experimental data from the literature [31].

$$
\begin{aligned}
G_m^{C15} = {} & y_{Al}^{I} y_{Al}^{II} G_{Al:Al}^{C15} + y_{Al}^{I} y_{Er}^{II} G_{Al:Er}^{C15} + y_{Er}^{I} y_{Al}^{II} G_{Er:Al}^{C15} + y_{Er}^{I} y_{Er}^{II} G_{Er:Er}^{C15} \\
& + 2RT\,(y_{Al}^{I} \ln y_{Al}^{I} + y_{Er}^{I} \ln y_{Er}^{I}) + RT\,(y_{Al}^{II} \ln y_{Al}^{II} + y_{Er}^{II} \ln y_{Er}^{II}) \\
& + y_{Al}^{I} y_{B}^{I} (y_{Al}^{II} \sum_n {}^{n}L_{Al,Er:Al}^{C15} (y_{Al}^{I} - y_{Er}^{I})^n + y_{Er}^{II} \sum_n {}^{n}L_{Al,Er:Er}^{C15} (y_{Al}^{I} - y_{Er}^{I})^n) \\
& + y_{Al}^{II} y_{Er}^{II} (y_{Al}^{I} \sum_n {}^{n}L_{Al:Al,Er}^{C15} (y_{Al}^{II} - y_{Er}^{II})^n + y_{Er}^{I} \sum_n {}^{n}L_{Er:Al,Er}^{C15} (y_{Al}^{II} - y_{Er}^{II})^n)
\end{aligned}
\tag{8}
$$

where y_i^I (i=Al or Er) and y_j^{II} (j=Al or Er) are the site fractions of components i and j in the first and second sublattices, respectively. The parameter $G_{i:j}^{C15}$ represents the Gibbs energy of the stable or metastable compounds in which the first and second sublattices are occupied by i and j, respectively, which are relative to the enthalpies of pure Al and Er in their SER states. ${}^{n}L_{Al,Er:j}^{C15}$ represents the nth interaction parameter between Al and Er in the first sublattice, while the second sublattice is occupied by the component j. ${}^{n}L_{i:Al,Er}^{C15}$ represents the nth interaction parameter between Al and Er in the second sublattice, while the first sublattice is occupied by the component i.

3.5. Intermetallic compounds $L1_2$ and solution phase fcc(A1)

In order to describe the structural relationship between the solution phase fcc(A1) (disordered part) and the phase $L1_2$ ($cP4$-$AuCu_3$-type, ordered part), the two phases fcc(A1) and $L1_2$ were modeled by $(Al,B,Er)_1(Va)_1$ and $(Al,B,Er)_{0.75}(Al,B,Er)_{0.25}(Va)_1$. Here, the symbol "Va" represents a vacancy. The Gibbs energy per

mole of formula unit $(Al,B,Er)_{0.75}(Al,B,Er)_{0.25}(Va)_1$ is described as

$$
\begin{aligned}
G_m^{L1_2} = {} & y_{Al}^{I} y_{Al}^{II} y_{Va}^{III} G^{L1_2}_{Al:Al:Va} + y_{Al}^{I} y_{B}^{II} y_{Va}^{III} G^{L1_2}_{Al:B:Va} + y_{Al}^{I} y_{Er}^{II} y_{Va}^{III} G^{L1_2}_{Al:Er:Va} \\
& + y_{B}^{I} y_{Al}^{II} y_{Va}^{III} G^{L1_2}_{B:Al:Va} + y_{B}^{I} y_{B}^{II} y_{Va}^{III} G^{L1_2}_{B:B:Va} \\
& + y_{B}^{I} y_{Er}^{II} y_{Va}^{III} G^{L1_2}_{B:Er:Va} + y_{Er}^{I} y_{Al}^{II} y_{Va}^{III} G^{L1_2}_{Er:Al:Va} + y_{Er}^{I} y_{B}^{II} y_{Va}^{III} G^{L1_2}_{Er:B:Va} \\
& + y_{Er}^{I} y_{Er}^{II} y_{Va}^{III} G^{L1_2}_{Er:Er:Va} + 0.75RT\,(y_{Al}^{I} \ln y_{Al}^{I} + y_{B}^{I} \ln y_{B}^{I} + y_{Er}^{I} \ln y_{Er}^{I}) \\
& + 0.25RT\,(y_{Al}^{II} \ln y_{Al}^{II} + y_{B}^{II} \ln y_{B}^{II} + y_{Er}^{II} \ln y_{Er}^{II}) \\
& + RT\,(y_{Va}^{III} \ln y_{Va}^{III}) + y_{Al}^{I} y_{B}^{I} (y_{Al}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,B:\,Al:Va} (y_{Al}^{I} - y_{B}^{I})^n \\
& + y_{B}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,B:\,B:Va} (y_{Al}^{I} - y_{B}^{I})^n \\
& + y_{Er}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,B:\,Er:Va} (y_{Al}^{I} - y_{B}^{I})^n) \\
& + y_{Al}^{I} y_{Er}^{I} (y_{Al}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,Er:\,Al:Va} (y_{Al}^{I} - y_{Er}^{I})^n \\
& + y_{B}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,Er:\,B:Va} (y_{Al}^{I} - y_{Er}^{I})^n \\
& + y_{Er}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al,Er:\,Er:Va} (y_{Al}^{I} - y_{Er}^{I})^n) \\
& + y_{B}^{I} y_{Er}^{I} (y_{Al}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B,Er:\,Al:Va} (y_{B}^{I} - y_{Er}^{I})^n \\
& + y_{B}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B,Er:\,B:Va} (y_{B}^{I} - y_{Er}^{I})^n \\
& + y_{Er}^{II} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B,Er:\,Er:Va} (y_{B}^{I} - y_{Er}^{I})^n) \\
& + y_{Al}^{II} y_{B}^{II} (y_{Al}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al:Al,\,B:Va} (y_{Al}^{II} - y_{B}^{II})^n \\
& + y_{B}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B:Al,\,B:Va} (y_{Al}^{II} - y_{B}^{II})^n \\
& + y_{Er}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Er:Al,\,B:Va} (y_{Al}^{II} - y_{B}^{II})^n) \\
& + y_{Al}^{II} y_{Er}^{II} (y_{Al}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al:Al,\,Er:Va} (y_{Al}^{II} - y_{Er}^{II})^n \\
& + y_{B}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B:Al,\,Er:Va} (y_{Al}^{II} - y_{Er}^{II})^n \\
& + y_{Er}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Er:Al,\,Er:Va} (y_{Al}^{II} - y_{Er}^{II})^n) \\
& + y_{B}^{II} y_{Er}^{II} (y_{Al}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{Al:B,\,Er:Va} (y_{B}^{II} - y_{Er}^{II})^n, \\
& y_{B}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B:B,\,Er:Va} (y_{B}^{II} - y_{Er}^{II})^n \\
& + y_{B}^{I} y_{Va}^{III} \sum_n {}^{n}L^{L1_2}_{B:B,\,Er:Va} (y_{B}^{II} - y_{Er}^{II})^n)
\end{aligned}
\tag{9}
$$

where y_i^I (i=Al, B or Er), y_j^I (j=Al, B or Er), and y_{Va}^{III} are the site fractions of components i , j and Va in the first, second and third sublattices, respectively. The parameters $G^{L1_2}_{i:\,j:\,Va}$ represent the Gibbs energies of the stable or metastable compounds, with the first, the second and the third sublattices are occupied by i, j and Va, respectively, which are relative to the enthalpies of pure Al, B ${}^{n}L^{L1_2}_{Al,B:i:Va}$, ${}^{n}L^{L1_2}_{Al,Er:i:Va}$, and ${}^{n}L^{L1_2}_{B,Er:i:Va}$ are the nth interaction parameters between (Al and B), (Al and Er), and (B and Er) in the first sublattice, while the second and the third sublattices are occupied by the components j and Va, respectively. ${}^{n}L^{L1_2}_{i:Al,B:Va}$, ${}^{n}L^{L1_2}_{i:Al,Er:Va}$ and ${}^{n}L^{L1_2}_{i:\,B,\,Er:\,Va}$ are the nth interaction parameters between (Al and B), (Al and Er), and (B and Er) in the second sublattice, while the first and the third sublattices are occupied by the components i and Va, respectively.

4. Assessment procedure

A general rule for selection of the adjustable parameters is that only those coefficients whose values were determined by experiment can be adjusted [41]. The assessment was carried out by means of the optimization module PARROT of the thermodynamic software Thermo-Calc [42], which can handle various kinds of experimental information.

The thermodynamic modeling and descriptions of the Al–B and B–Er systems were carefully revised in order to assess the ternary Al–B–Er system. The thermodynamic parameters of the ternary Al–B–Er system were optimized on the basis of experimental data available in the literature [35–37]. The details of the present assessment are given in Section 5.

5. Results and discussions

The thermodynamic description of the Al–B–Er system obtained in this study is listed in Table 2. Fig. 2 presents the calculated Al–B phase diagram using the experimental data obtained from the literature [9,12,43,44]. The miscibility gap in the liquid at high temperatures does not appear until 10,000 K in this calculated phase diagram. In the study of Mirković et al. [9], the solubility of B in the fcc(A1) phase was described by a two-sublattice model $(Al)_1(Va,B)_1$ in the Al–B system, after comparing with the solubility of B in the fcc(A1) phase in B–Ni and other systems. However, limit direct experimental data were available to support the two-sublattice model $(Al)_1(Va,B)_1$. The order-disorder transition between the solution phase fcc(A1) and the intermetallic compound $L1_2$ cannot be built during the evaluation of the Al–B–Er system if the solution phase fcc(A1) is described by the two-sublattice model $(Al)_1(Va,B)_1$ in the Al–B system [9]. In order to build the order–disorder transition in the present study, the model of the solution phase fcc(A1) in the Al–B system was described by the substitutional solution model $(Al,B)_1(Va)_1$. Table 3 lists the invariant reactions of the Al–B system. Fig. 3 represents the calculated enthalpies of formation of the compounds in the Al–B system at 298 K, compared with the experimental data from the literature [45–48]. The enthalpies of formation of AlB_{12} in Mechel and Kleppa [45] and Kant and Moon [46] were found to be more reliable than the experimental results obtained by Domalski and Armstrong [47,48]. Mirković et al. [9] explained this by observing that the starting materials were more reliable in former studies, leading to more accurate experimental results. On the other hand, the phase equilibria of the isothermal sections in the Al–B–Er ternary system cannot be built correctly by the parameters in Mirković et al.'s study; hence, in order to obtain the correct phase equilibria of isothermal sections in the Al–B–Er ternary system, the Gibbs energies of the intermetallic compounds AlB_2 and AlB_{12} found therein may be corrected by the addition of small positive values.

Fig. 4 represents the B–Er phase diagram calculated using the experimental data from Liao and Spear [31]. The temperature of the miscibility gap for this calculated phase diagram is ~6602 K, which is an acceptable result. Table 4 lists the invariant reactions of the B–Er system. Fig. 5 represents the calculated enthalpies of formation at 298 K in the B–Er system using both the experimental data [32] and the predicted values [33] from Meschel and Kleppa. The present calculated phase equilibria of the B–Er system are similar to those of Wang et al. [10], however, the Gibbs energies of the four intermetallic compounds in Wang et al.'s study may also be corrected with the addition of large negative values in order to obtain the correct phase equilibria of the isothermal sections in the Al–B–Er ternary system.

Figs. 6, 7, 8 and 9 represent the calculated isothermal sections of the Al–B–Er system, at 1873 K, 1073 K, 873 K and 298 K, respectively, in the entire composition range calculated using the present thermodynamic description.

As can be seen in Fig. 6, the six three-phase regions (triangle symbol) and three two-phase regions (circle symbol) are in good

Table 4
Invariant reactions of the B–Er system.

Reaction	Type	Present work				Experimental data [31]			
		T(K)		x(Er)		T(K)		x(Er)	
liq. → $B_{66}Er$ + rhom	Eutectic	2340	7.61×10^{-3}	–	1.000	2293	0.010	–	1.00
liq. → $B_{66}Er$	Congruent	2343	–	–		2343	–	–	
liq. → $B_{66}Er + B_{12}Er$	Eutectic	2325	4.00×10^{-2}	–	–	2288	0.030	–	–
liq. + B_4Er → $B_{12}Er$	Eutectic	2356	7.41×10^{-2}	–	–	2356	0.070	–	–
liq. → B_4Er	Congruent	2633	–	–		2633	–	–	
liq. + B_4Er → B_2Er	Peritectic	2458	0.356	–	–	2458	0.350	–	–
liq. → B_2Er + hcp	Eutectic	1773	0.975	–	0.999	1773	0.990	–	1.00

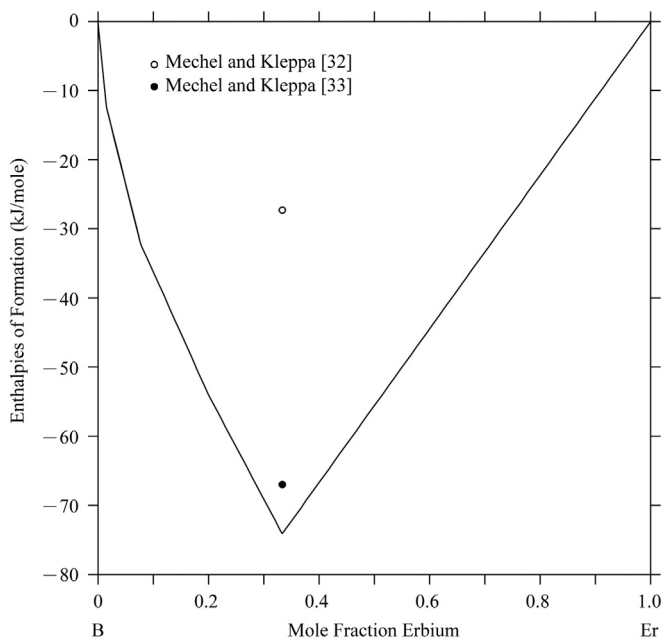

Fig. 5. Enthalpies of formation at 298 K for the B–Er system, calculated using experimental data from the literature [32,33].

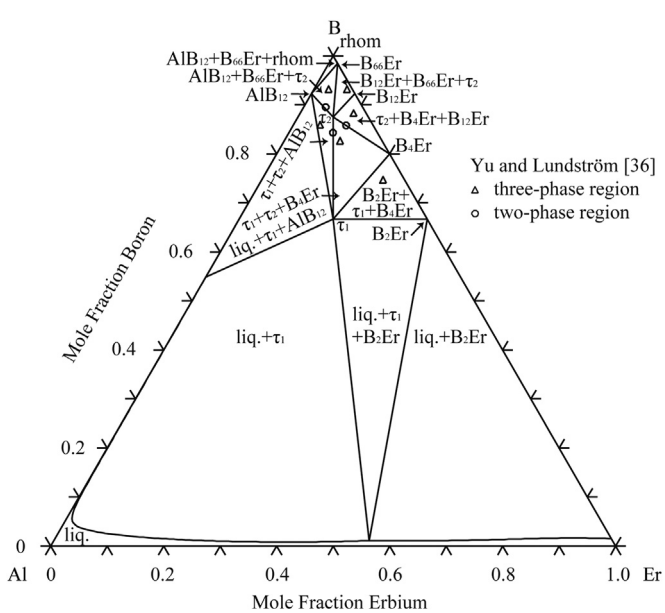

Fig. 6. Isothermal section of the Al–B–Er system at 1873 K, calculated using the present thermodynamic description, with experimental data obtained from the literature [36].

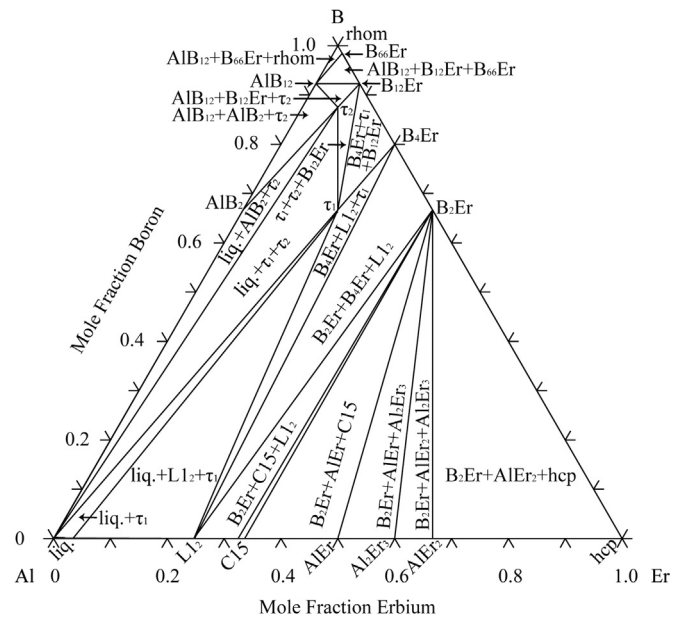

Fig. 7. Isothermal section of the Al–B–Er system at 1073 K, calculated using the present thermodynamic description.

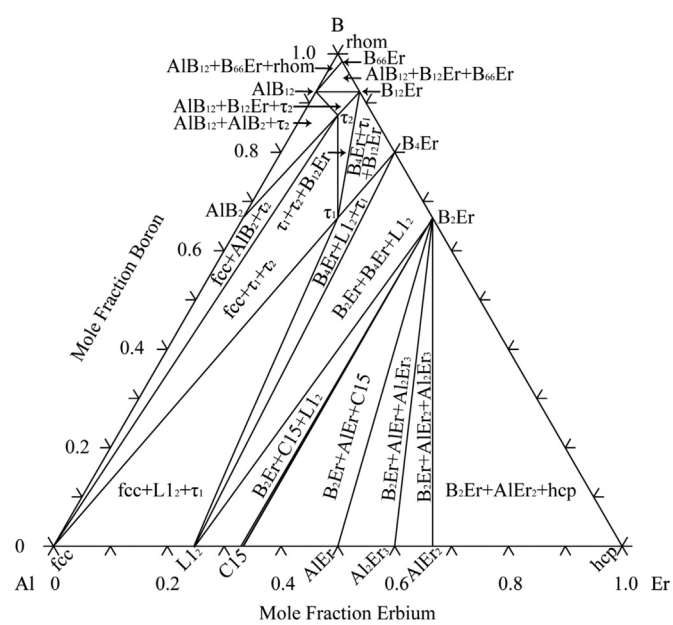

Fig. 8. Isothermal section of the Al–B–Er system at 873 K, predicted using the present thermodynamic description.

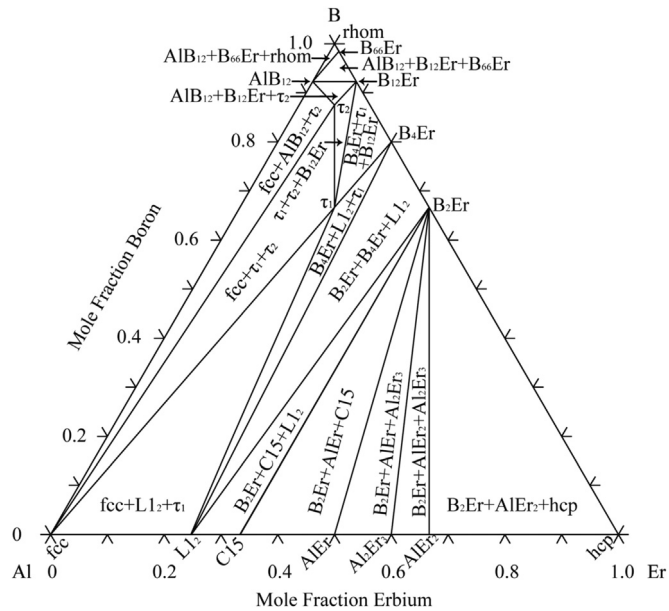

Fig. 9. Isothermal section of the Al–B–Er system at 298 K, predicted using the present thermodynamic description.

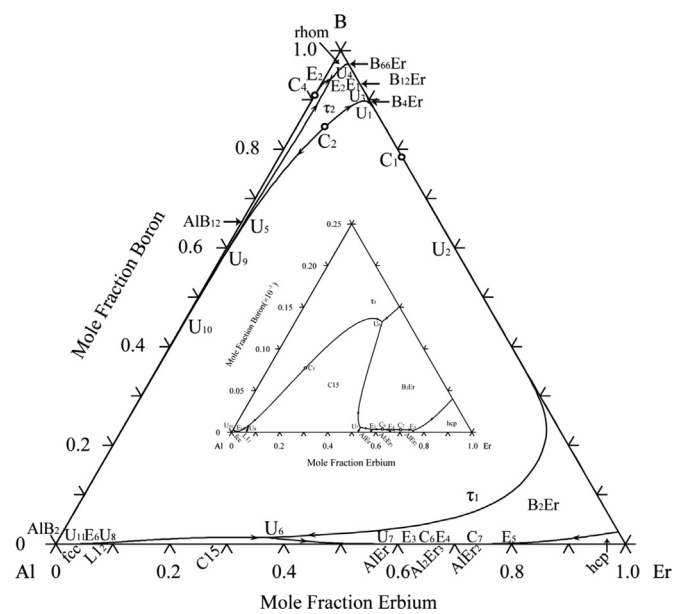

Fig. 10. Projection of the liquidus surfaces in the Al–B–Er system, calculated using the present thermodynamic description.

agreement with the experimental data in the composition range of ~66–100 at% B from Yu and Lundström [36].

Raghavan [49] reviewed the isothermal sections of the Al–B–Er system at 1873 K and 1073 K based on the experimental data in Chaban et al. [35] and Yu and Lundström [36]. However, for the isothermal section at 1073 K, the three-phase regions $B_2Er+C15+L1_2$ and $B_2Er+B_4Er+L1_2$ in the study of Chaban et al. [35] were changed to $B_4Er+C15+L1_2$ and $B_2Er+B_4Er+C15$ in the study of Raghavan [49]. Therefore, in Fig. 7, the phase equilibria of the calculated isothermal section at 1073 K is based on Chaban et al.'s experimental data [35].

As the τ_1 and τ_2 phases were not found by Petsow and Effenberg in their 1990 study [37], they evaluated the isothermal section at 873 K in the composition range of 0–75 at% B, excluding these two phases. Figs. 8 and 9 represent the predicted isothermal sections for this system at 873 K and 298 K calculated using the present thermodynamic description, based on data from Chaban et al. [35] and Yu and Lundström [36]; hence, the results shown in Figs. 7, 8 and 9 have similar phase equilibria. Due to the phase equilibria of the Al–B and Al–Er binary systems, the main differences in the isothermal sections for these systems at 1073 K, 873 K and 298 K were in the composition range of 0–0.25 at% Er: four three-phase regions liquid$+AlB_2+\tau_2$, $AlB_{12}+AlB_2+\tau_2$, liquid$+\tau_2+\tau_2$ and liquid$+L1_2+\tau_1$ in Fig. 7, four three-phase

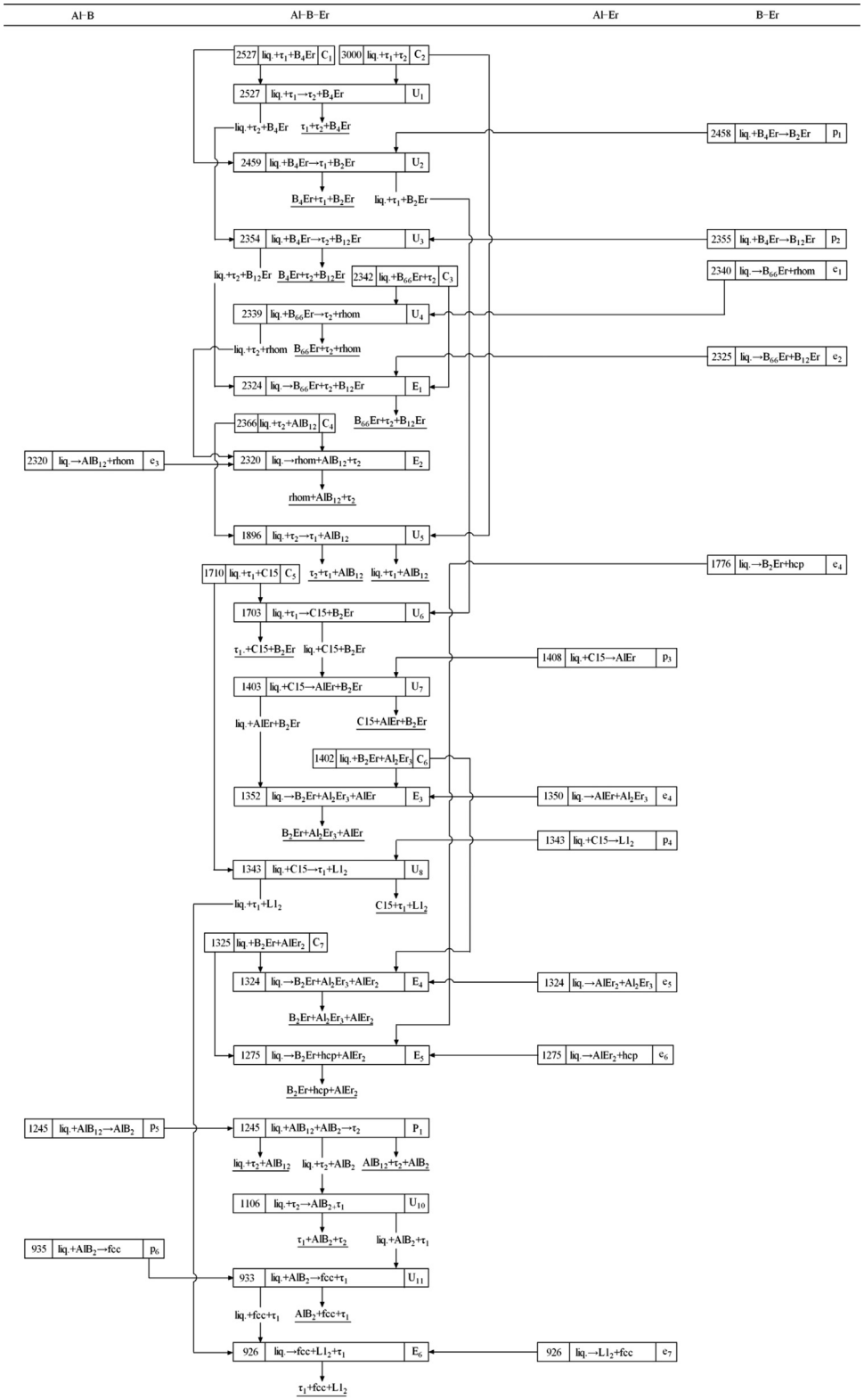

Fig. 11. Predicted reaction scheme for the Al–B–Er system.

regions fcc+AlB_2+τ_2, AlB_{12}+AlB_2+τ_2, fcc+τ_2+τ_2 and fcc+ $L1_2$+τ_1 in Fig. 8, and three three-phase regions fcc+AlB_{12}+τ_2, fcc+τ_2+τ_2 and fcc+ $L1_2$+τ_1 in Fig. 9.

Fig. 10 represents the predicted projection of the liquidus surfaces of the Al–B–Er system using the present thermodynamic description. Fig. 11 represents the predicted reaction scheme in the Al–B–Er system based on our calculated results. Table 5 lists the invariant reactions in the Al–B–Er system.

6. Conclusion

A thermodynamic description of the Al–B–Er system was

Table 5
Invariant reactions for the Al–B–Er system.

Reaction	Present work	
	Type	T(K)
liq. + τ_1 + B_4Er	C_1	2527
liq. + τ_1 + τ_2	C_2	3000
liq. + τ_2 + $B_{66}Er$	C_3	2342
liq. + τ_2 + AlB_{12}	C_4	2366
liq. + τ_1 + C15	C_5	1710
liq. + Al_2Er_3 + B_2Er	C_6	1402
liq. + $AlEr_2$ + B_2Er	C_7	1325
liq. + τ_1 → B_4Er + τ_2	U_1	2527
liq. + B_4Er → τ_1 + B_2Er	U_2	2459
liq. + B_4Er → τ_2 + $B_{12}Er$	U_3	2354
liq. + $B_{66}Er$ → τ_2 + rhom	U_4	2339
liq. → τ_2 + $B_{12}Er$ + $B_{66}Er$	E_1	2324
liq. → τ_2 + AlB_{12} + rhom	E_2	2320
liq. + τ_2 → AlB_{12} + τ_1	U_5	1896
liq. + τ_1 → C15 + B_2Er	U_6	1703
liq. + C15 → $AlEr$ + B_2Er	U_7	1403
liq. → B_2Er + $AlEr$ + Al_2Er_3	E_3	1352
liq. + C15 → τ_1 + $L1_2$	U_8	1343
liq. → $AlEr_2$ + B_2Er + Al_2Er_3	E_4	1324
liq. → $AlEr_2$ + B_2Er + hcp	E_5	1275
liq. + AlB_{12} + AlB_2 → τ_2	P_1	1245
liq. + τ_2 → τ_1 + AlB_2	U_9	1106
liq. + AlB_2 → fcc + τ_1	U_{10}	933
liq. → $L1_2$ + fcc + τ_1	E_6	926

evaluated on the basis of the experimental data obtained from the literature. A set of self-consistent thermodynamic parameters for describing the Gibbs energy of individual phases as functions of composition and temperature are provided. The isothermal sections, the predicted projection of the liquidus surfaces and the reaction scheme of the Al–B–Er system have also been provided. The theoretical calculations made using the thermodynamic description from this study showed good agreement with experimental data taken from the literature.

Acknowledgments

Special thanks to Professor Z.M. Du, C.R. Li and C.P. Guo at the University of Science and Technology Beijing.

This work was supported by the Science and Technology Development Project of Jilin Provincial Science and Technology Department (Grant nos. 20140520100JH and 20140204043GX), the Science and Technology Development Project of Jilin Municipal Science and Technology Bureau (Grant no. 201467008), and the Research Fund for the Doctoral Program of Northeast Dianli University (Grant no. BSJXM-201224).

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